Input system:
Dependent variables:x
Independent variables:t
Equations:

$$\frac{dx}{dt} = bx^2(K - x)\text{inv}(K(N + x))$$

==========

Result:
Dependent variables:s_1, x
Independent variables:t
Additional variables:

$$s_1 = \text{inv}(K(N + x))$$

Equations:

$$\frac{dx}{dt} = s_1 bx^2 K + (-1)s_1 bx^3$$

$$\frac{ds_1}{dt} = (-1)s_1^3 K^2 bx^2 + s_1^3 Kbx^3$$